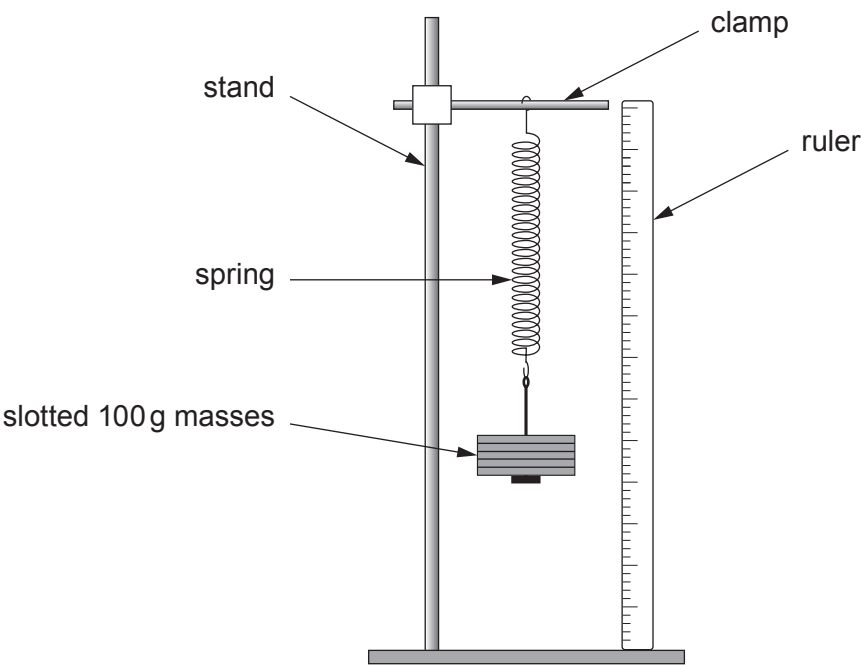


4. (a) Describe how the apparatus shown below is used to investigate the extension of a spring. [6 QER]



- Include the following in your answer:
- how the apparatus is set up,
 - a description of the method used to obtain the results,
 - how the results are analysed.

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(b) A spring is 3 cm long. When an object is added to it, its length increases to 18 cm.

(i) Calculate the extension produced by the object. [1]

extension = cm

(ii) The spring constant of the spring is 0.8 N/cm.
Use an equation from page 2 to calculate the force produced by the object on the spring. [2]

force = N

(iii) This force is the weight of the object.
Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the object. (Gravitational field strength, $g = 10 \text{ N/kg}$) [2]

mass = kg

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